BlueHive Self-Update System

How a build reaches a television. Reference for operators and maintainers.

1 Delivery flow

Three independent channels. None depends on the others; each is published by its own switch on `release.ps1`.

Channel	Serves	Consumed by	Switch
VPS	updates.daigurrenblue.com	The app's own self-updater. This is what fielded TVs check.	-Vps
GitHub	releases/tag/v<code>	OGD add-on install + update. Requires a version.json asset or OGD installs once and never updates.	-GitHub
LAN	192.168.1.136:8090	Local development only. Written to data\ on the build machine.	(default)

Publish → **check** → **download** → **install.** The client polls the manifest, compares `latest_version_code` against its own `BuildConfig.VERSION_CODE`, downloads the APK, verifies SHA-256, and hands it to `PackageInstaller`. The install always requires a user press on a system dialog — a sideloaded app cannot install silently.

2 Manifest (/bluehive/version)

Field	Meaning / effect
latest_version_code	Integer. An update is offered when this is **strictly greater** than the installed code.
latest_version_name	Display only.
apk_url	Only the **filename** is used — the client prepends its own UPDATE_BASE_URL. A wrong host is harmless; a wrong filename is fatal.
sha256	Verified after download. Lowercase hex. A blank value makes the client refuse the update as unverifiable.
min_supported	Builds below this may not decline. 0 = never mandatory. **release.ps1 -Vps sets 0 unless -Mandatory.**
release_notes	Display only.

Must be UTF-8 **without a BOM**. A BOM makes Android's org.json throw, the client reports "no update" forever, and nothing logs why.

3 Parameters

Constant	Value	Where	Notes
callTimeout / connect / read	4 s	UpdateManager.kt	Manifest fetch. Blocks the launch, so this **is** the blank-window duration. 8 s was measured worse.
connect / read (download)	30 s / 120 s	UpdateManager.kt	APK transfer, ~165 MB.
UPDATE_CHECK_INTERVAL_MS	6 h	BlueHiveApplication.kt	Throttle for the foreground-return check.
KeepReleases	3	release.ps1	Local APK pruning. The VPS is not pruned.
CHECK_FLOOR_MS	3 s	HostEntitlementRelay.kt	Entitlement re-check floor (reference pattern).
SECOND_PASS_DELAY_MS	6 s	HostEntitlementRelay.kt	Corrective second read after a cold host process.

4 Files and responsibilities

File	Responsibility
update/UpdateManager.kt	checkForUpdate() · isMandatory() · downloadAndVerify() · install(). Owns the HTTP clients and the SHA-256 gate.
update/SelfUpdateActivity.kt	The update screen. Checks install permission **before** downloading, then downloads, verifies and commits. Blocks Back.
update/SelfUpdateInstallReceiver.kt	PackageInstaller callbacks. Launches the system confirm dialog on PENDING_USER_ACTION.
update/PackageReplacedReceiver.kt	Relaunches after the process is replaced.
update/ServerConfig.kt	updateBaseUrl, from BuildConfig.UPDATE_BASE_URL. Trims trailing slashes.
auth/AuthActivity.kt	Launch-path check. Routes to the updater, LoadingScreen, or sign-in.
BlueHiveApplication.kt	Foreground-return check, ForceUpdateBus handler, single-flight launchUpdater().
bluehive-updates/release.ps1	Publishes all three channels. Hashes, uploads, writes the manifest, verifies the served bytes.
bluehive_plugin/device_sessions.py	POST /api/android/admin/force-update-check — pushes force_update over SSE.

5 Operator commands

`.\release.ps1 -Vps -GitHub -Push`	Full release: publish both public channels, then ask live devices to re-check.
`.\release.ps1 -Vps`	Publish only. Devices pick it up next launch, or within 6 h on a foreground return.
`-Mandatory`	Sets min_supported = this build. Devices below it are held on the update screen with no way into the app.

`BH_ADMIN_KEY`	Env var read by -Push. The endpoint answers 503 when no key is configured — disabled by default.

The push carries **no version, URL or hash** — only "re-read the manifest". The event travels a channel the TV does not control, so anything it asserted would be a way to install arbitrary APKs on every television at once.

6 Known limitations

Detection is unreliable	The manifest fetch blocks the launch behind a 4 s budget. Measured 4 of 6 cold launches on an onn 4K box. A miss is silent: the device stays on the old build. The fix is to take the check off the launch path — attempted and reverted, needs its own pass.
Install needs a press, always	PackageInstaller shows a system dialog on every update. Only a privileged installer (e.g. OGD, which already holds REQUEST_INSTALL_PACKAGES) can avoid it.
Permission is per-device, per-app	REQUEST_INSTALL_PACKAGES cannot be pre-granted. On some Google TV builds the Settings entry is unreachable — the in-app prompt's Settings button is the only route.
Interrupts mid-session	ForceUpdateBus has no playback guard. A -Push while someone is watching drops them onto the update screen.
Release logging is stripped	R8 removes Log.v/d/i/w. Only Log.e survives, so use it for anything needed to diagnose a shipped build.